PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
from collections import deque
places = ["door", "window", "middle"]
start = ("door", "window", False, False)
def next_states(state):
    monkey, box, on_box, has = state
    if not on_box:
        for p in places:
            if p != monkey:
                yield f"Walk from {monkey} to {p}", (p, box, False, has)
        if monkey == box:
            for p in places:
                if p != box:
                    yield f"Push box from {box} to {p}", (p, p, False, has)
            yield "Climb on the box", (monkey, box, True, has)
    else:
        yield "Climb down from the box", (monkey, box, False, has)
        if monkey == "middle":
            yield "Grab the banana", (monkey, box, True, True)
def solve():
    queue = deque([(start, [])])
    visited = {start}
    while queue:
        state, actions = queue.popleft()
        if state[3]:
            return actions
        for action, nxt in next_states(state):
            if nxt not in visited:
                visited.add(nxt)
                queue.append((nxt, actions + [action]))
plan = solve()
print("Optimal action plan:")
for i, step in enumerate(plan, 1):
    print(f"  {i}. {step}")
```

OUTPUT:

```
>>>
========== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ==========
Optimal action plan:
  1. Walk from door to window
  2. Push box from window to middle
  3. Climb on the box
  4. Grab the banana
>>>
```

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